B 本文所用全部提示词内容

B.1 分子名称与 SMILES 代码的双向转换

提示词 1

```
What is the SMILES code for {molecule}? Only output the SMILES string, with
    no additional text or explanation.
```

提示词 2

```
{
  "task_type": "name_to_smiles",
  "input_data": {
    "molecule_name": "{molecule}"
  },
  "output_format": "Return only the SMILES string as plain text, with no
    other content."
}
```

提示词 3

```
{
  "role": "You are an expert chemist.",
  "task": "Provide the SMILES string for the given chemical name. Your
    response must contain only the SMILES string itself, without any
    additional text or explanation.",
  "molecule_name": "{molecule}"
}
```

提示词 4

```
{
  "role": "You are an expert chemist specializing in chemical informatics.
    Your task is to provide the accurate ca-nonical SMILES string.",
  "instruction": "The system prompt contains examples. Follow them precisely.
     Your output must be ONLY the SMILES string.",
  "query": {
    "name": "{molecule}"
  }
}
```

提示词 5

```
What is the name of the molecule for the SMILES code {smiles_code}? Only
    output the molecule name, with no ad-ditional text or explanation.
```

提示词 6

```
{
  "task_type": "smiles_to_name",
  "input_data": {
    "smiles_code": "{smiles_code}"
```

```
  },
  "output_format": "Return only the molecule name as plain text, with no
    other content."
}
```

提示词 7

```
{
  "role": "You are an expert chemist.",
  "task": "Provide the common or IUPAC name for the given SMILES string. Your
     response must contain only the molecule name itself, without any
    additional text or explanation.",
  "smiles_code": "{smiles_code}"
}
```

提示词 8

```
{
  "role": "You are an expert chemist specializing in chemical informatics.
    Your task is to provide the common or IUPAC name for a given SMILES
    string.",
  "instruction": "The system prompt contains examples. Follow them precisely.
     Your output must be ONLY the molecule name.",
  "query": {
    "smiles": "{smiles_code}"
  }
}
```

B.2 辛醇-水分配系数（log P）预测

提示词 1

```
What is the logP Value of {molecule}? Only output logP Value, with no
    additional text or explanation.
```

提示词 2

```
{
  "task_type": "Determine the logP Value",
  "input_data": {
    "molecule_name": "{molecule}"
  },
  "output_format": "Return only the molecular logP Value as plain text, with
    no other content."
}
```

提示词 3

```
{
  "role": "You are an expert chemist specializing in perdict logP value",
  "task": "Provide the logP value of the molecule for the given chemical name
```

```
    . Your response must contain only the logP value itself, without any
    additional text or explanation.",
  "molecule_name": "{molecule}"
}
```

B.3 极性表面积（PSA）预测

提示词 1

```
What is the Polar Surface Area value of {molecule}? Only output Polar Surface
     Area Value, with no additional text or explanation.
```

提示词 2

```
{
  "task_type": "Determine the Polar Surface Area",
  "input_data": {
    "molecule_name": "{molecule}"
  },
  "output_format": "Return only the molecular Polar Surface Area Value as
    plain text, with no other content."
}
```

提示词 3

```
{
  "role": "You are an expert chemist specializing in perdict Polar Surface
    Area",
  "task": "Provide the Polar Surface Area value of the molecule for the given
     chemical name. Your response must contain only the Polar Surface Area
    value itself, without any additional text or explanation.",
  "molecule_name": "{molecule}"
}
```

B.4 分子几何构型的判定

提示词 1

```
What is the Molecular geometry of {molecule} according to VSEPR theory? Only
    output the Molecular geometry name, with no additional text or
    explanation.
```

提示词 2

```
{
  "task_type": "Determine the molecular geometry based on the VSEPR theory",
  "input_data": {
    "molecule_name": "{molecule}"
  },
```

```
  "output_format": "Return only the molecular geometry name as plain text,
   with no other content."
}
```

提示词 3

```
{
  "role": "You are an expert chemist specializing in Molecular geometry and
   VSEPR theory",
  "task": "Provide the Molecular geometry name for the given chemical name.
   Your response must contain only the Molecular geometry name itself,
   without any additional text or explanation.",
  "molecule_name": "{molecule}"
}
```

提示词 4

```
{
  "role": "You are an expert chemist specializing in Molecular geometry and
   VSEPR theory. Your task is to provide the accurate canonical Molecular
   geometry name.",
  "instruction": "The system prompt contains examples. Follow them precisely.
     Your output must be ONLY the Molecular geometry name.",
  "query": {
    "name": "{molecule}"
  }
}
```

B.5 分子点群的判定

提示词 1

```
What is the point group of {molecule}? Only output the point group name, with
     no additional text or explanation.
```

提示词 2

```
{"task_type": "Identify the molecular point group name",
  "input_data": {"molecule_name": "{molecule}"},
  "output_format": "Return only the point group name as plain text, with no
   other content.}
```

提示词 3

```
{
  "role": "You are an expert chemist specializing in structural chemistry",
  "task": "Provide the point group name for the given chemical name. Your
   response must contain only the point group name itself, without any
   additional text or explanation.",
  "molecule_name": "{molecule}"
}
```

提示词 4

```
{
  "role": "You are an expert chemist specializing in structural chemistry.
   Your task is to provide the accurate canonical point group name.",
  "instruction": "The system prompt contains examples. Follow them precisely.
    Your output must be ONLY the point group name.",
  "query": {
    "name": "{molecule}"
  }
}
```

B.6 ^{1}H NMR 谱图信号数量的预测

提示词 1

```
How many groups of peaks will {molecule} exhibit in the 1H NMR spectrum? Only
    output the number of peak groups, with no additional text or explanation
   .
```

提示词 2

```
{
  "task_type": "Determine the number of peak groups in 1H NMR spectrum",
  "input_data": {
    "molecule_name": "{molecule}"
  },
  "output_format": "Return only the number of peak groups as plain text, with
    no other content."
}
```

提示词 3

```
{
  "role": "You are an expert chemist specializing in NMR spectroscopy",
  "task": "Provide the number of peak groups of the molecule for the given
   chemical name. Your response must contain only the number of peak groups
   value itself, without any additional text or explanation.",
  "molecule_name": "{molecule}"
}
```

提示词 4

```
{
  "role": "You are an expert chemist specializing in NMR spectroscopy. Your
   task is to provide the number of peak groups in 1H NMR spectrum for a
   given molecule.",
  "instruction": "The system prompt contains examples. Follow them precisely.
    Your output must be ONLY the number of peak groups.",
  "query": {
    "molecule_name": "{molecule}"
  }
```

```
}
```

B.7 XRD 谱图特征峰位置的预测

提示词 1

```
How many peaks will a well-crystallized {molecule} material exhibit in the
    XRD pattern using Cu αK radiation source (test range 10°-80°)? Output
    only the positions of the peaks, without any additional text or
    explanation.
```

提示词 2

```
{
  "task_type": "Determine the positions of peaks in XRD spectrum (test range
    10°-80°) using a Cu αK radiation source (Assume that the material has
    good crystallinity)",
  "input_data": {
    "molecule_name": "{molecule}"
  },
  "output_format": "Return only the positions of the peaks as plain text,
    with no other content."
}
```

提示词 3

```
{
  "role": "You are an expert chemist specializing in X-ray diffraction",
  "task": "Provide the positions of peaks for the given material (test range
    10°-80°). Your response must contain only the peaks itself, without any
    additional text or explanation.",
  "molecule_name": "{molecule}"
}
```

提示词 4

```
{
  "role": "You are an expert chemist specializing in X-ray diffraction. Your
    task is to provide the positions of the peaks for a given material (test
    range 10°-80°).",
  "instruction": "The system prompt contains examples. Follow them precisely.
     Your output must be ONLY the positions of the peaks.",
  "query": {
    "molecule_name": "{molecule}"
  }
}
```

B.8 晶体学信息文件（CIF）的模拟与生成测试

提示词 1

```
Generate the cif file of the {molecule}, and output only the complete content
    of the cif file without additional text or explanation.
```

提示词 2

```
{
  "task_type": "Generate cif file",
  "input_data": {
    "molecule_name": "{molecule}"
  },
  "output_format": "Return only the complete content of the cif file, with no
    other content."
}
```

提示词 3

```
{
  "role": "You are an expert chemist specializing in generate cif file",
  "task": "the complete content of the cif file for the given material or
   molecule. Your response must contain only the complete content of the cif
    file, without any additional text or explanation.",
  "molecule_name": "{molecule}"
}
```

提示词 4

```
{
  "role": "You are an expert chemist specializing in generate cif file. Your
   task is to provide the complete con-tent of the cif file for a given
   material or explanation.",
  "instruction": "The system prompt contains examples. Follow them precisely.
    Your output must be ONLY the complete content of the cif file, without
   any additional text or explanation.",
  "query": {
    "molecule_name": "{molecule}"
  }
}
```